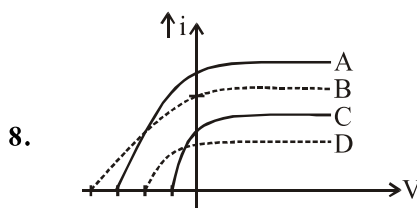


DUAL NATURE OF RADIATION & MATTER

1. The photo electrons emitted from a surface of sodium metal are such that they:
 - (1) All are of the same frequency
 - (2) Have the same KE
 - (3) Have the same de-Broglie wave-length
 - (4) Have their speed varying from zero to a certain maximum
2. The wave-length of the matter-wave depends on :
 - (1) Mass
 - (2) Velocity
 - (3) Momentum
 - (4) All of these
3. In a photoelectric experiment, if stopping potential is applied, the photo current becomes zero. This means that:
 - (1) The emission of photoelectron is stopped
 - (2) The photoelectrons are emitted but are accumulated near the emitter metal surface.
 - (3) The photoelectrons are dispersed from the emitter surface
 - (4) None
4. If the frequency of light in a PEE is doubled then maximum KE of photoelectron:
 - (1) Doubled
 - (2) Be halved
 - (3) Become more than double
 - (4) Become less than double
5. Let P and E denote the linear momentum and energy of photon. (in same medium) for another photon of smaller wavelength:
 - (1) Both P and E increase
 - (2) P increase and E decrease
 - (3) P decrease and E increase
 - (4) Both P and E decrease
6. Stopping potential for photoelectrons:
 - (1) Does not depend on the frequency of incident light
 - (2) Does not depend on the nature of cathode material
 - (3) Depends on both the frequency of incident light and nature of cathode material
 - (4) Depends on intensity of incident light

7. If an electron and a photon propagate in form of waves having the same wavelength, it implies that they can have the same:
 - (1) Energy
 - (2) Momentum
 - (3) Velocity
 - (4) Angular momentum

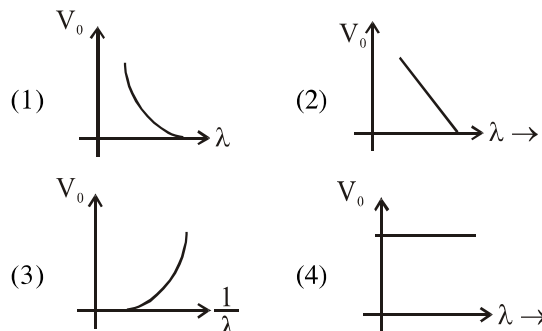


The figure shows the result of an experiment involving photoelectric effect.

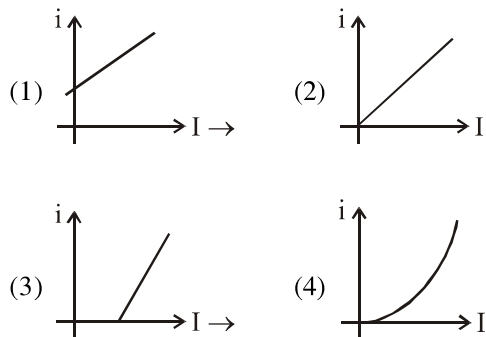
The graph A, B, C and D relate to a light beam having different wavelength.

Select correct:

- (a) Beam B has a highest frequency
 - (b) Beam C has longest wavelength
 - (c) Beam A has highest rate of photoelectric emission
 - (d) Highest momentum is possessed by some electrons emitted by B
- (1) a & b only
 - (2) a, b & c
 - (3) a, b, c & d
 - (4) a & c only
9. For photo-electric effect,
 $\lambda \rightarrow$ incident photon wavelength
 $V_0 \rightarrow$ stopping potential.

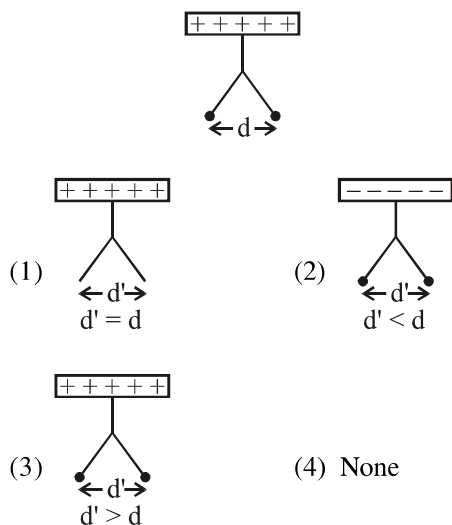


10. The graph between intensity of light falling on a metallic plate (I) with current (i) generated is:



11. A polychromatic light is used in an experiment on photoelectric effect. The stopping potential:
- Is related to mean wave-length
 - Is related to the longest wave-length
 - Is related to shortest wave-length
 - Is not related to the wave-length
12. The equation $E = pc$ is valid:
- For an electron as well as for photon
 - For an electron but not for a photon
 - For a photon but not for electron
 - Neither for an electron nor a photon
13. Which of the properties should be in metal for thermionic emission:
- Low melting point and high work function
 - High melting point and low work function
 - High melting point and high work function
 - Low melting point and low work function
14. Two Black object of same diameter, a sphere and a disc are placed in front of uniform beam of light. The plane of disc is perpendicular to light rays. The force acting upon them by the light is:
- Zero
 - Bigger on disc
 - Bigger on sphere
 - The same on both
15. In photoelectric effect, if a weak intensity radiation instead of strong intensity of suitable frequency is used then:
- Photoelectric effect will get delayed
 - Photoelectric effect will not take place
 - Maximum KE will decrease (for photo electron)
 - Saturation current will decrease
16. Free electrons can not normally escape out of metal surface because:
- Free electrons are bounded to one or other ion core
 - Free electrons lie deep inside atom
 - Free electron are held in metal lattice by attraction force of protons
 - If an electron attempts to come out of metal, the metal surface acquires a positive charge and pulls the electron back to metal
17. Colour of glow in a discharge tube at a pressure of 0.001 mm of mercury column depends on:
- Nature of gas in the discharge tube
 - Potential difference between cathode and anode
 - Nature of material of cathode
 - Nature of glass of the discharge tube
18. X-ray can be deflected by applying:
- A magnetic field
 - An electric field
 - Both (1) and (2)
 - None of these
19. Value of $\frac{e}{m}$ (specific charge) of the cathode particles:
- Depends on potential difference of cathode and anode
 - Depends on nature of gas in the discharge tube
 - Is independent of material of cathode or gas in tube
 - Depends on nature of metal used for cathode
20. Lenard observed that no electrons are emitted when frequency of light is less than a certain minimum frequency. This minimum frequency depends on:
- Potential difference of emitter and collector plates
 - Distance between collector and the emitter plate
 - Size (area) of the emitter plate
 - Material of the emitter plate
21. Photo emission with visible light is possible in:
- alkali metals
 - alkaline earth metals
 - metalloids
 - Semiconductors

22. A positively charged zinc plate is connected to an electroscope. It is then irradiated by UV light. Result is:



23. Stopping potential is more negative for:
- Higher frequency of incident radiation
 - Lower frequency of incident radiation
 - Higher intensity of incident radiation
 - Lower intensity of incident radiation
24. In an electron-photon collision, which is not true?
- Photon-electron collision is elastic
 - Photon can rebound from an electron
 - In an photon electron collision number of photons is not conserved
 - Photons are deflected by a strong electric field
25. Macroscopic objects in our daily life do not show wave like properties:
- de-Broglie's hypothesis is not true
 - de-Broglie's hypothesis is true only for electron
 - de-Broglie's hypothesis is true only for photons
 - Wavelength associated with macroscopic object is small
26. Wave nature of electrons is exploited in:
- mass spectrometer
 - coolidge tube
 - synchrotrons
 - electron microscope
27. Wavelength of an electromagnetic wave is:
- more than de-Broglie wavelength of its photon
 - less than de-Broglie wavelength of its photon
 - there is no relation between wavelength of a radiation and de-Broglie wavelength of its photon
 - wavelength of radiation is equal to de-Broglie of its photon

28. Cathode rays are:
- Streams of positive ions
 - Streams of negative charged particles
 - Streams of nuclei
 - Streams of neutrons
29. Cathode rays are streams of fast moving negative charged particles. Their speed range (consider $c \approx 3 \times 10^8$ m/s):
- $0.1 c$ to $0.2 c$
 - c
 - Greater than c
 - Around $10^{-5}c$ to $10^{-6}c$
30. Photoemission of electrons:
- is instantaneous process with a time lag of less than nano second
 - is slow with a large time lag
 - shows a time lag of around 1 second
 - None of these
31. The stopping potential for having zero photo-electron current:
- is inversely porportional to wavelength of incident light
 - increases uniformly with increase in wavelength of incident light
 - is not dependent on frequency of incident light
 - increases linearly with increase in frequency of incident light
32. Work function depends on:
- frequency of incident light
 - properties of metal
 - Intensity of incident light
 - Nature of surface of metal
- (1) (a, b) (2) (c, d) (3) (b, d) (4) (b, c)
33. The time taken by photo e^- to come out after photon strikes is approximately:
- (1) 10^{-9} s (2) 10^{-4} s (3) 10^{-1} s (4) 1 s
34. Which of the following is sometimes called electric eye:
- Zener diode
 - LED
 - Photo cell
 - Solar cell
35. Work function is maximum for:
- Aluminium
 - Sodium
 - Caesium
 - Platinum
36. In photon-particle collision which of following quantities may not be conserved:
- Total energy
 - Number of photons
 - Total momentum
 - None of these
37. Caesium is sensitive to:
- Ultra violet rays
 - Visible light
 - Both (1) and (2)
 - None

38. G.P. Thomson experimentally confirmed existence of matter waves of electrons by phenomena:
 (1) Refraction (2) Diffraction
 (3) Polarisation (4) Reflection
39. Which of the following statements are true:
 (a) Dual nature of matter is evident in de-Broglie equation $\lambda = \frac{h}{p}$.
 (b) Wave nature of macroscopic particles are not visible in daily life
 (c) $\lambda = \frac{h}{p}$ is not satisfied by a photon
 (1) (a, c) (2) (a, b)
 (3) (b, c) (4) (a, b, c)
40. In Davisson-Germer experiment, deflection in ammeter connected to e^- detector is proportional to:
 (1) Kinetic energy of electrons entering detector
 (2) Momentum of electrons entering detector
 (3) Intensity of electron beam entering detector
 (4) None of these
41. Which of the following statements is correct regarding photons:
 (1) Photons do not carry momentum
 (2) Photon is a material particle
 (3) Photon energy depends on intensity of given light
 (4) In interaction of radiation with matter radiations behaves like photons
42. Resolving power of electron microscope is:
 (1) less than optical microscope
 (2) better than optical microscope
 (3) equal to optical microscope
 (4) none of these
43. Minimum frequency of incident light required to produce photo electric effect depends on:
 (1) distance between light source and metal
 (2) Intensity of light
 (3) Stopping potential
 (4) Nature of metal
44. The photocurrent in experiment of photo electric effect increases if:
 (1) intensity of source is increased
 (2) exposure time is increased
 (3) exposure time is decreased
 (4) intensity of source is decreased
45. Which phenomenon best supports the theory that matter has a wave nature ?
 (1) Electron momentum
 (2) Electron diffraction
 (3) Photon momentum
 (4) Photon
46. The specific charge of an electron is :-
 (1) 1.6×10^{-19} coulomb
 (2) 4.8×10^{-10} stat coulomb
 (3) 1.76×10^{11} coulomb/kg
 (4) 1.76×10^{-11} coulomb/kg
47. Energy conversion in photoelectric cell takes place from :-
 (1) Chemical to electrical
 (2) Magnetic to electrical
 (3) Optical to electrical
 (4) Mechanical to electrical
48. The electron are emitted in the photoelectric effect from a metal surface :-
 (1) Only if the frequency of the incident radiation is above a certain threshold value
 (2) Only if the temperature of the surface is high
 (3) At a rate that is independent of the nature of the metal
 (4) With a maximum velocity proportional to the frequency of the incident radiation
49. The order of photo current that can be obtained from photo cell is :-
 (1) nA (2) mA
 (3) μ A (4) None of these